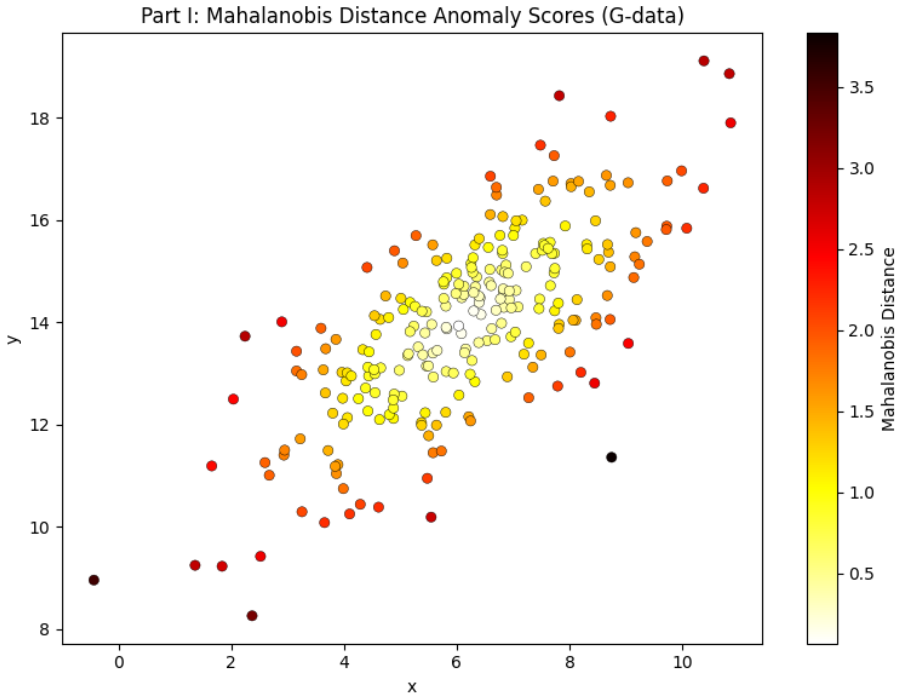
Lab 9: Anomaly Detection in Data

By: Abdullah Mohammed

100784442

CSCI 4150U

1 Part I



Index	x	y	mahal
41	8.747885	11.359886	3.831225
8	-0.438606	8.958312	3.528900
26	2.368932	8.259790	3.225480
67	2.243288	13.728437	2.862487
204	10.386685	19.115480	2.856695

Table 1: Top 5 Anomalies (Mahalanobis)

The Mahalanobis distance results show that the detected anomalies are points that lie far from the center of the overall distribution. The scatterplot shows most points concentrated around the middle of the diagonal trend, while the top anomalies, such as indices 41, 8, and 26, appear farther from this central pattern. This means Mahalanobis distance is effective here because it does not just measure straight-line distance, but also how unusual a point is relative to the full structure of the dataset.

2 Part II

2.1 compound Dataset

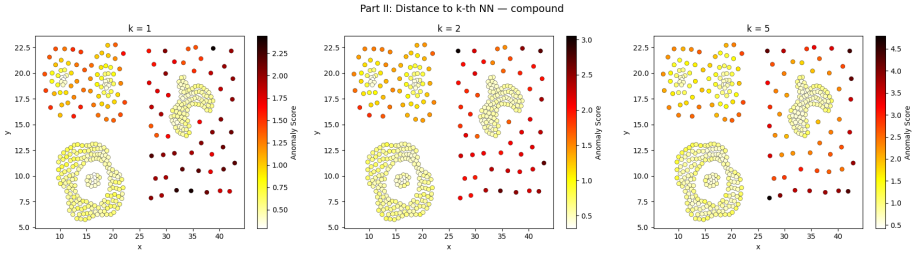


Table 2: Top 5 Anomalies for Compound Data at Different k Values

k	Index	x	y	Score
1	43	38.8500	22.4000	2.4418
1	22	34.7000	8.5500	2.4005
1	21	31.9500	8.6000	2.3265
1	13	27.2500	11.9500	2.2023
1	36	42.9000	11.2500	2.1840
2	0	26.7500	22.1500	3.0500
2	42	42.2500	22.1500	2.7166
2	36	42.9000	11.2500	2.6912
2	22	34.7000	8.5500	2.6707
2	43	38.8500	22.4000	2.5080
5	19	27.1500	7.8500	4.7885
5	42	42.2500	22.1500	4.4096
5	37	41.9500	8.5000	4.3012
5	0	26.7500	22.1500	4.2154
5	40	42.5500	19.4500	4.1863

2.2 flame Dataset

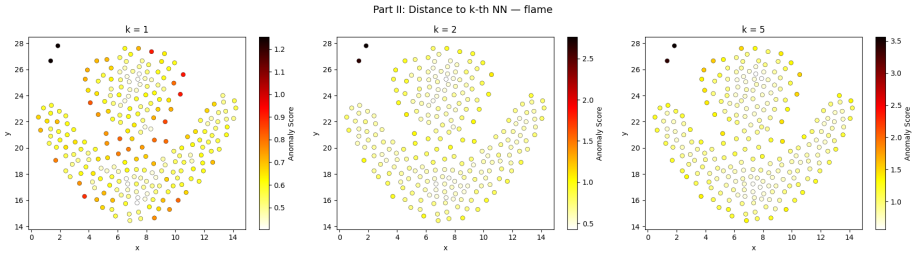


Table 3: Part II: Top 5 Anomalies for Flame Data at Different k Values

k	Index	x	y	Score
1	0	1.8500	27.8000	1.2540
1	1	1.3500	26.6500	1.2540
1	204	10.5500	25.6000	0.9192
1	192	10.3500	24.1000	0.9179
1	201	8.3500	27.3500	0.9124
2	0	1.8500	27.8000	2.7613
2	1	1.3500	26.6500	2.6443
2	204	10.5500	25.6000	1.0817
2	168	4.6500	22.5500	1.0548
2	163	5.2000	21.1000	1.0308
5	0	1.8500	27.8000	3.5584
5	1	1.3500	26.6500	3.4004
5	197	4.5500	26.6500	1.5532
5	196	3.9000	25.9500	1.5508
5	204	10.5500	25.6000	1.5133

2.3 pathbased Dataset

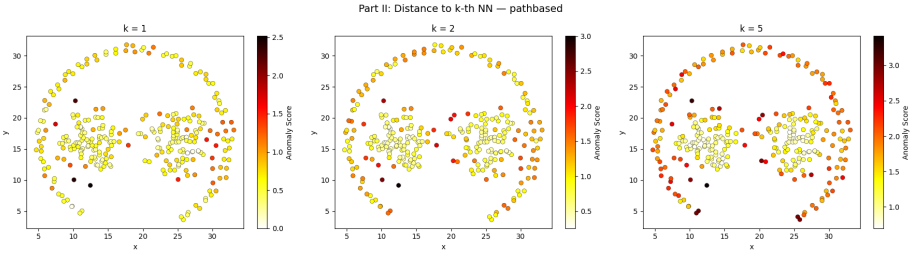


Table 4: Part II: Top 5 Anomalies for Pathbased Data at Different k Values

k	Index	x	y	Score
1	202	12.4500	9.1500	2.5164
1	203	10.1000	10.0500	2.4135
1	270	10.3000	22.7500	2.2771
1	266	7.4500	19.0000	1.7205
1	106	29.1500	16.5000	1.5338
2	202	12.4500	9.1500	3.0017
2	203	10.1000	10.0500	2.5164
2	270	10.3000	22.7500	2.3265
2	294	17.9000	15.6000	2.1360
2	197	19.9500	19.8000	2.0555
5	202	12.4500	9.1500	3.4154
5	270	10.3000	22.7500	3.2442
5	0	11.2500	5.0500	3.0270
5	1	10.9500	4.7000	2.9833
5	102	25.5000	4.1000	2.9741

For the k-NN distance method, the results show that the detected anomalies are usually points that are more isolated from the rest of the data. In the compound dataset, many of the top anomalies are points located near the edges of clusters or in areas with fewer nearby neighbors. In the flame dataset, points such as indices 0 and 1 appear repeatedly for different k values, showing that they are consistently far from the main flame-shaped pattern. In the pathbased dataset, indices like 202 and 270 also appear multiple times, which suggests these points are in the ends of paths. As k increases, the anomaly scores also increase, which makes sense because the method is comparing each point to a larger neighborhood.

3 Part III

3.1 Method 1: Density = 1 / (distance to k-th neighbor)

3.1.1 compound dataset

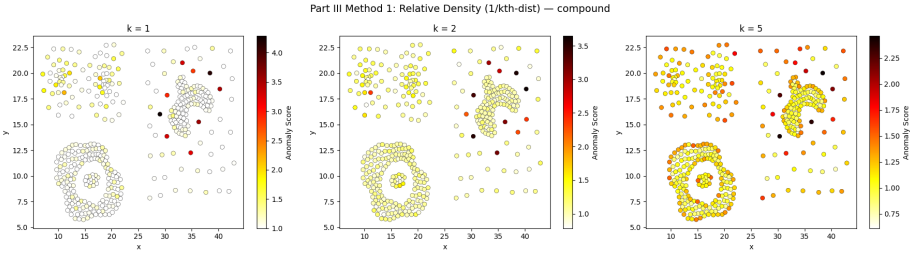


Table 5: Method 1: Density = 1/ (distance to k-th neighbor), Compound Data

k	Index	x	y	Score
1	9	29.0500	16.0000	4.2851
1	44	38.4000	20.0000	4.1548
1	24	36.3000	15.2500	3.5553
1	39	40.2500	18.4500	3.4819
1	46	33.2500	21.0000	3.2855
2	39	40.2500	18.4500	3.6449
2	44	38.4000	20.0000	3.5738
2	12	30.3500	13.8500	3.5287
2	7	30.4000	17.8500	3.3924
2	23	34.8000	12.2500	3.1943
5	44	38.4000	20.0000	2.4608
5	12	30.3500	13.8500	2.4293
5	24	36.3000	15.2500	2.3715
5	7	30.4000	17.8500	2.2517
5	39	40.2500	18.4500	2.1189

3.1.2 flame dataset

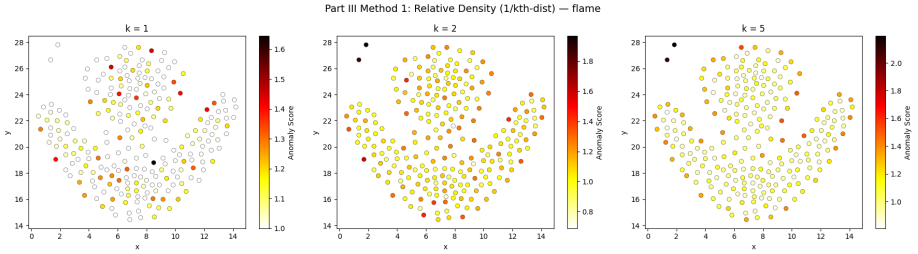


Table 6: Method 1: Density = 1/ (distance to k-th neighbor), Flame Data

k	Index	x	y	Score
1	105	8.5000	18.8000	1.6440
1	225	5.5500	26.1000	1.4552
1	201	8.3500	27.3500	1.4427
1	139	12.2000	22.8500	1.4221
1	192	10.3500	24.1000	1.4080
2	0	1.8500	27.8000	1.9675
2	1	1.3500	26.6500	1.8629
2	20	1.7000	19.0500	1.5988
2	55	5.8500	14.8000	1.4418
2	223	4.6500	25.1000	1.4280
5	0	1.8500	27.8000	2.1935
5	1	1.3500	26.6500	2.1018
5	199	6.5000	27.6000	1.5458
5	174	9.3000	22.0000	1.5220
5	192	10.3500	24.1000	1.4961

3.1.3 pathbased dataset

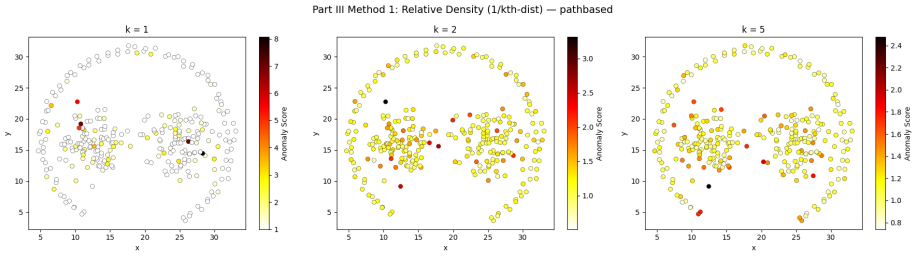


Table 7: Method 1: Density = 1/ (distance to k-th neighbor), Pathbased Data

k	Index	x	y	Score
1	116	28.3000	14.4500	8.0623
1	157	26.2500	16.3500	7.1063
1	265	10.8000	19.2000	6.9642
1	270	10.3000	22.7500	5.5227
1	264	10.5500	18.5500	5.0000
2	270	10.3000	22.7500	3.3151
2	294	17.9000	15.6000	2.6161
2	202	12.4500	9.1500	2.4948
2	293	16.6000	16.1500	2.3055
2	197	19.9500	19.8000	2.1647
5	202	12.4500	9.1500	2.4760
5	144	20.3000	13.1000	1.9393
5	111	27.4500	10.8500	1.8179
5	0	11.2500	5.0500	1.7654
5	1	10.9500	4.7000	1.7371

In Method 1, density is based on the distance to the k-th nearest neighbor. This means that points in low density areas are more likely to be considered anomalies. For the compound dataset, the detected anomalies are mostly points near the outside of the clusters, where there are fewer neighboring points. For the flame dataset, the method highlights points near the edges of the flame shape. For the pathbased dataset, the anomalies are often points near the ends of the paths or in areas where the data is spread out more. This shows that Method 1 is useful for finding points that are locally isolated.

3.2 Method 2: Density = 1 / (avg distance to k neighbors)

3.2.1 compound dataset

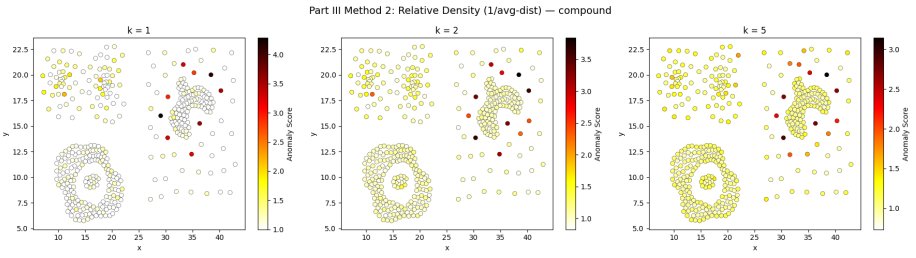


Table 8: Method 2: Density = 1/ (average distance to k neighbors), Compound Data

k	Index	x	y	Score
1	9	29.0500	16.0000	4.2851
1	44	38.4000	20.0000	4.1548
1	24	36.3000	15.2500	3.5553
1	39	40.2500	18.4500	3.4819
1	46	33.2500	21.0000	3.2855
2	44	38.4000	20.0000	3.8366
2	12	30.3500	13.8500	3.6279
2	39	40.2500	18.4500	3.5922
2	7	30.4000	17.8500	3.3751
2	24	36.3000	15.2500	3.3155
5	44	38.4000	20.0000	3.1433
5	12	30.3500	13.8500	2.9066
5	24	36.3000	15.2500	2.7432
5	7	30.4000	17.8500	2.7159
5	39	40.2500	18.4500	2.6705

“

3.2.2 flame dataset

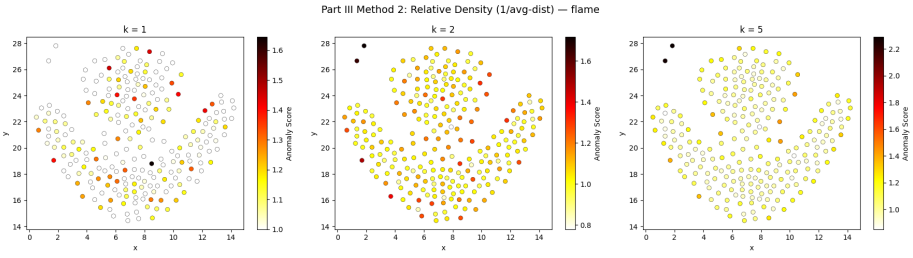


Table 9: Method 2: Density = 1/ (average distance to k neighbors), Flame Data

k	Index	x	y	Score
1	105	8.5000	18.8000	1.6440
1	225	5.5500	26.1000	1.4552
1	201	8.3500	27.3500	1.4427
1	139	12.2000	22.8500	1.4221
1	192	10.3500	24.1000	1.4080
2	0	1.8500	27.8000	1.7164
2	1	1.3500	26.6500	1.6518
2	20	1.7000	19.0500	1.4890
2	40	3.7000	16.3000	1.3885
2	132	11.7500	22.1000	1.3132
5	0	1.8500	27.8000	2.2894
5	1	1.3500	26.6500	2.2717
5	174	9.3000	22.0000	1.4402
5	158	9.3500	20.5000	1.4234
5	192	10.3500	24.1000	1.4196

3.2.3 pathbased dataset

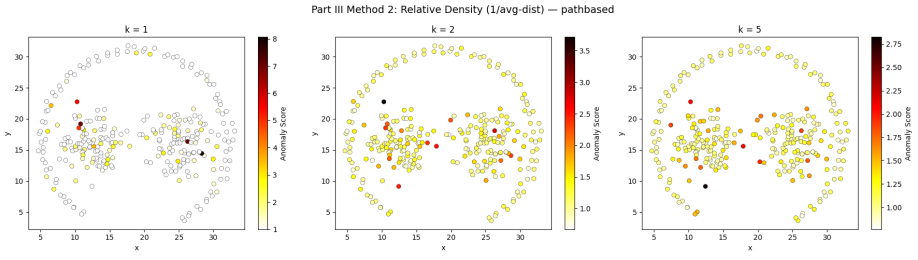


Table 10: Method 2: Density = 1/ (average distance to k neighbors), Pathbased Data

k	Index	x	y	Score
1	116	28.3000	14.4500	8.0623
1	157	26.2500	16.3500	7.1063
1	265	10.8000	19.2000	6.9642
1	270	10.3000	22.7500	5.5227
1	264	10.5500	18.5500	5.0000
2	270	10.3000	22.7500	3.7144
2	162	26.3000	18.1000	2.7866
2	294	17.9000	15.6000	2.5685
2	202	12.4500	9.1500	2.4932
2	264	10.5500	18.5500	2.4153
5	202	12.4500	9.1500	2.8257
5	270	10.3000	22.7500	2.0957
5	294	17.9000	15.6000	2.0781
5	144	20.3000	13.1000	1.9492
5	162	26.3000	18.1000	1.8977

In Method 2, density is based on the average distance to the k nearest neighbors instead of just one neighbor. Because of this, the results are a bit smoother and less sensitive to sudden changes. The same general types of points are still being detected as anomalies, especially points in sparse or outer regions, but the rankings are more stable. This makes Method 2 a little more reliable when looking at the overall neighborhood around a point, rather than depending too much on one single distance.